

Jonathan Woo

Education

University of California, Los Angeles
Doctor of Philosophy in Mathematics

August 2025 — June 2030 (anticipated)

University of California, Los Angeles
Bachelor of Science in Mathematics of Computation

September 2020 — August 2024
Cumulative GPA: 3.878/4.00 — Major GPA: 3.923/4.00

Publications

1. Wing Pok Lee*, **Jonathan D. Woo***, Luke F. Triplett*, Yifan Gu*, Sarah C. Burnett, Lingyun Ding, Andrea L. Bertozzi, A comparative study of dynamic models for gravity-driven particle-laden flows, *Applied Mathematics Letters*, Volume 164, 2025, 109480, ISSN 0893-9659, <https://doi.org/10.1016/j.aml.2025.109480>. (*equal contribution)

Posters and Presentations

Gravity-driven Particle-laden Free Surface Flow - A Comparison of Models
UCLA Undergraduate Research and Creativity Showcase

May 2024

- Presented comparisons between a diffusive flux model and suspension balance model in the context of thin-films.
- Numerical simulation data from each model reveal that both models agree well with experimental data and that the two models minimally differ from each other.

Phase transitions in highly concentrated particle-liquid thin films
76th Annual Meeting of the Division of Fluid Dynamics

November 2023

- Experimentally investigated phenomenon in gravity-driven particle-laden flows down an incline where liquid-particle suspensions transition from fluid-like behavior to solid-like behavior.
- Discovered quantitative dependence of front speed and fluid layer thickness on parameters such as the inclination angle, particle diameter, particle volume fraction, densities, and viscosity.

Modeling polydisperse particle-laden flow down an incline
76th Annual Meeting of the Division of Fluid Dynamics

November 2023

- Modelled behavior of particle-laden flows with finitely many particle species of differing size as well as a continuous distribution of particle sizes.
- Developed model consisting of a system of hyperbolic conservation laws whose fluxes were determined by an auxiliary ordinary differential equation system (for the finite species case) or an integro-differential equation (for the continuous size distribution case).
- Numerically simulated and performed comparisons between physical experimental data and numerical data.

Research Experience

Graduate Student Researcher

Los Alamos National Laboratory, Advancing Machine Learning for Scientific Discovery Fellowship

May 2026 — Present

- Studying probabilistic modal reconstruction and time series forecasting from sparse sensors with uncertainty quantification.

Undergraduate Researcher

UCLA Computational and Applied Mathematics REU

June 2023 — August 2023

Mentors: Andrea Bertozzi, Sarah Burnett, Lingyun Ding

- Studied gravity-driven particle-laden viscous thin-films down an incline through physical experiments, computational simulations, data analysis, and theoretical exploration.
- Explored and compared continuum models for transport of particles and liquid derived from a diffusive flux model and a suspension balance model

Teaching Experience

Teaching Assistant

Department of Mathematics, UCLA

- Differential and Integral Calculus (Math 31A) Fall 2025
- Integration and Infinite Series (Math 31B) Winter 2026
- Machine Learning (Math 156) Spring 2026

Projects

Computer Graphics Class Project

October 2022 — December 2022

Department of Computer Science, UCLA

- With two other group members, built an interactive computer graphics demonstration found at <https://bruinkart.glitch.me/>.
- Implemented 3D graphics with lighting, shading, models, and physics in JavaScript.

Work Experience

Math Tutor

Mathanisum Learning Center

January 2025 — July 2025

- Provided exceptional instruction/tutoring services to students.
- Evaluate and correct student work and homework.
- Interact and motivate students.
- Work collaboratively with team members to deliver the best learning experience for students.

PIC Lab Assistant

UCLA Programming in Computing Lab

September 2023 — June 2024

- Maintained proper operations of computing lab through cleaning, organizing, and assisting.
- Assisted lab patrons in troubleshooting and programming.